Neurogenic Orthostatic Hypotension Module — Mechanism Research Report

Module slug: `neurogenic_orthostatic_hypotension` · **Category:** Module **Prepared:** 2026-11-05 · Literature-based curation (no primary data files)

Summary

Neurogenic orthostatic hypotension (nOH) is best modeled as a convergent, multi-entry directed acyclic graph (DAG). Four mechanistically distinct lesion sites all funnel into a single required conformance target — **deficient norepinephrine (NE) release at the sympathetic neurovascular junction on standing** — which in turn drives failure of baroreflex-mediated vasoconstriction and an orthostatic fall in blood pressure with cerebral and muscle hypoperfusion. This convergence is the strongest-supported shared mechanism in the module and is confirmed by two independent reviews that state verbatim that all nOH disorders "are characterized by defective norepinephrine release from sympathetic terminals upon standing" (PMID: 28702747) and that NE deficiency can arise peripherally (denervation) or centrally (deficient sympathetic traffic to intact terminals) (PMID: 26373628).

The four lesion branches are gene- and disease-specific entry points, not independent modules: (a) **central/preganglionic degeneration** (multiple system atrophy, MSA) with postganglionic neurons spared and normal plasma NE; (b) **ganglionic nicotinic transmission failure** (autoimmune autonomic ganglionopathy, AAG) caused by anti-ganglionic acetylcholine receptor [α3/CHRNA3] autoantibodies, with both neurons surviving; (c) **postganglionic noradrenergic denervation** (pure autonomic failure [PAF], Parkinson disease [PD], dementia with Lewy bodies [DLB]) with low plasma NE; and (d) **NE-synthesis failure** inside intact neurons (dopamine β-hydroxylase [DBH] deficiency; CYB561 deficiency). Three of the four branches have direct experimental confirmation — the *Dbh*-knockout mouse for branch (d), a rabbit immunization model plus electrophysiology for branch (b), and quantitative intermediolateral-nucleus neuropathology for branch (a).

The lesion site is clinically and therapeutically decisive because it sets residual sympathetic reserve. Patients with peripheral denervation and low plasma NE respond better to "norepinephrine replacers" (midodrine, droxidopa), which bypass the missing transmitter at any lesion site; patients with preserved peripheral fibers but impaired central drive (MSA)

respond better to "norepinephrine enhancers" (atomoxetine, pyridostigmine), which amplify residual traffic and therefore require surviving efferent fibers (PMID: 32979782). Two optional side branches — adrenoceptor denervation supersensitivity (postganglionic lesions) and neurogenic supine hypertension with nocturnal pressure natriuresis — complete the module. Alpha-synuclein aggregation, reflex (vasovagal) syncope, and postural orthostatic tachycardia syndrome (POTS) are explicitly out of scope.

Executive Recommendation: Module Name and Scope

Retain **"Neurogenic Orthostatic Hypotension Module."** It is the field-standard term, used verbatim across multiple reviews (PMID: 32979782, PMID: 26373628, PMID: 28702747, PMID: 26092297), and the neurogenic/non-neurogenic boundary is explicitly drawn in the literature (PMID: 31733702). Model it as a **convergent multi-entry DAG**: conformance should require the shared NE-release-deficit backbone plus **≥1 lesion branch** (not a single exclusive branch), because branch overlap is real — IML preganglionic loss occurs in Lewy body disorders as well as MSA (PMID: 27061071).

Key Findings

F1 — Convergent mechanism: all lesion sites meet at deficient NE release

The single most important structural feature of this module is convergence. Two independent reviews establish the shared conformance target. Baker & Kimpinski state that the nOH disorders "are characterized by defective norepinephrine release from sympathetic terminals upon standing, resulting in impaired vasoconstriction" (PMID: 28702747). Loavenbruck & Sandroni localize the deficiency to two origins that nonetheless converge: "Norepinephrine deficiency in NOH can arise peripherally, due to cardiovascular sympathetic denervation (as in pure autonomic failure, Parkinson's disease, and a variety of neuropathies), or centrally, due to a failure of viscerosensory signals to generate adequate sympathetic traffic to intact sympathetic nerve endings (as in multiple system atrophy)" (PMID: 26373628). This is the backbone of the DAG: regardless of upstream lesion, the pathway passes through a shared NE-release deficit before producing the downstream hemodynamic phenotype. This node should be **core and required for conformance**.

F2 — Lesion site determines sympathetic reserve and which drug class works

The branching structure is not merely descriptive — it predicts treatment response. Park et al. show that patients "with peripheral noradrenergic degeneration (pure autonomic failure, Parkinson's disease) and low plasma norepinephrine, tend to respond better to 'norepinephrine replacers' (midodrine and droxidopa). On the other hand, patients with relatively preserved 'sympathetic reserve', i.e., those with impaired central autonomic pathways but spared peripheral noradrenergic fibers (multiple system atrophy) and normal or slightly reduced plasma norepinephrine, tend to respond better to 'norepinephrine enhancers' (pyridostigmine, atomoxetine, and yohimbine)" (PMID: 32979782). Okamoto et al. add that "even severely affected patients have residual sympathetic tone which can be harnessed for their treatment" (PMID: 30571543). This establishes the mechanistic rationale for keeping the lesion branches distinct within the DAG: they encode different amounts of residual sympathetic reserve, which is the pharmacological substrate.

F3 — Biochemical/imaging separation of preganglionic (MSA) from postganglionic (PAF/PD) lesions

The branches can be distinguished at the bench and bedside. Goldstein et al. found distinct plasma catechol profiles: "The PAF group had low NE, DHPG, NMN, EPI, and MN levels, the group with MSA + NOH had generally normal levels" (PMID: 12707437). Clement et al. summarize the anatomical distinction and its imaging correlate: "In PD and PAF, pathogenic involvement is mainly post-ganglionic, while in MSA, the involvement is mainly pre-ganglionic," with MIBG myocardial scintigraphy measuring postganglionic fiber integrity and distinguishing PD from MSA (PMID: 38216420). A supporting physiological study (clonidine challenge) found a greater vasodepressor effect in MSA than PAF, consistent with partial preservation of brainstem sympathetic outflow in MSA and greater peripheral α2-adrenoceptor denervation supersensitivity in PAF (PMID: 16404729). Plasma NE and cardiac sympathetic imaging are therefore the practical biomarkers that assign a patient to a branch.

F4 — Ganglionic transmission-failure branch (b): anti-gAChR antibodies, both neurons spared

Branch (b) is biologically distinct because the neurons survive and the deficit is in synaptic transmission. Goldstein et al. describe "AAG reflecting decreased ganglionic neurotransmission due to circulating antibodies to the neuronal nicotinic receptor and PAF being a Lewy body

disease with prominent loss of sympathetic noradrenergic nerves" (PMID: 19155193). The antibody is a diagnostic marker: "High levels of anti-ganglionic nicotinic acetylcholine receptor (gAChR) serum antibodies are detected in approximately 50% of AAG cases, confirming the diagnosis" (PMID: 40900869). This branch is also the one most amenable to immunotherapy rather than pressor pharmacology, underscoring why it must remain a distinct entry point.

F5 — Branch (b) molecular identity and causal proof: α3/CHRNA3 subunit, rabbit model, pleiotropy

The molecular target and causal chain for branch (b) are unusually well pinned down. Vernino et al. report that "The ganglionic (alpha3-type) neuronal AChR mediates fast synaptic transmission in sympathetic, parasympathetic and enteric autonomic ganglia… These antibodies recognize the alpha3 subunit of the ganglionic AChR, and thus do not bind non-specifically to other nicotinic AChR… Antibodies from patients with AAG inhibit ganglionic AChR currents and impair transmission in autonomic ganglia. An animal model of AAG in the rabbit recapitulates the important clinical features of the human disease" (PMID: 18951069). Pechlivanidou et al. independently identify "the α3 subunit containing nAChR (α3-nAChR)… detected and likely pathogenic in Autoimmune Autonomic Ganglionopathy" (PMID: 37164280). Critically, the same antibody produces **multiple manifestations** — "orthostatic hypotension, gastrointestinal dysmotility, anhidrosis, bladder dysfunction and sicca symptoms" (PMID: 18951069) — direct human evidence that one causal factor yields pleiotropic autonomic phenotypes. This branch has the fullest causal chain in the module: antibody → current inhibition (in vitro) → impaired transmission → autonomic failure (rabbit model + human).

F6 — Branch (d): DBH and CYB561 loss-of-function cause selective noradrenergic failure, rescued by droxidopa

Branch (d) is the "pure synthesis" lesion, with intact neurons that cannot make NE. Shibao et al. establish the biochemistry: "Cytochrome b561 (CYB561) generates ascorbic acid, a cofactor in the enzymatic conversion of dopamine to norepinephrine by dopamine β-hydroxylase," and "CYB561 deficiency is characterized by selective sympathetic noradrenergic failure with lifelong, disabling OH but with normal sympathetic cholinergic (sweating) and parasympathetic (heart rate regulation) functions" (PMID: 31822578). Arnold et al. describe the parent disorder: "Dopamine β-hydroxylase (DBH) deficiency is a rare genetic disorder characterized by failure to convert dopamine to norepinephrine" (PMID: 27778639). Van den

Berg et al. first implicated CYB561, finding low NE/EPI but normal DBH activity (PMID: 29343526). Both genes converge on the same enzymatic step and both phenotypes are correctable by droxidopa, which supplies NE downstream of the block.

F7 — Drug-target logic maps cleanly onto the branches

The module's pharmacology is a direct read-out of its mechanism. Kaufmann et al. describe the universal replacer: "Droxidopa, an orally active synthetic amino acid that is converted to norepinephrine by the enzyme aromatic L-amino acid decarboxylase (dopa-decarboxylase)... is presumed to raise BP by acting at the neurovascular junction to increase vascular tone" (PMID: 26092297). Okamoto et al. describe the reserve-dependent enhancers: "norepinephrine transporter blockade with atomoxetine raises blood pressure (BP) in autonomic failure patients by increasing synaptic norepinephrine concentrations; acetylcholinesterase inhibition with pyridostigmine increases BP by facilitating ganglionic cholinergic neurotransmission to increase sympathetic outflow" (PMID: 30571543). Jung et al. note atomoxetine's effects were "pronounced in cases with central autonomic failure, including multiple-system atrophy" (PMID: 36647224), while a 4-week RCT found atomoxetine was not superior to placebo on symptom score, flagging variable efficacy (PMID: 39294522). The pyridostigmine trial confirmed it improves standing BP without worsening supine hypertension (PMID: 16476804).

F8 — Side branch: adrenoceptor denervation supersensitivity

Postganglionic lesions sensitize vascular adrenoceptors. Biaggioni et al. showed that "Patients with autonomic failure, even those with severe sympathetic deprivation, are hypersensitive to the pressor effects of yohimbine. This increased responsiveness can be explained by sensitization of adrenergic receptors, analogous to denervation supersensitivity" plus loss of baroreflex buffering (PMID: 8089252). Park et al. reinforce that "Denervation hypersensitivity and impairment of baroreflex buffering makes these patients sensitive to small doses of pressor agents" (PMID: 32979782). This explains why low plasma NE predicts a larger pressor response to droxidopa and direct agonists, and it should be modeled as an **optional side branch specific to postganglionic lesions**.

F9 — Side branch: supine hypertension with nocturnal pressure natriuresis worsens morning OH

Roughly half of nOH patients have supine hypertension, which sets up a self-worsening loop. Jordan & Biaggioni state that "about one half of these patients also suffer from supine hypertension, which induces pressure natriuresis, worsening orthostatic hypotension," and

that this "is due to residual sympathetic tone in patients with multiple system atrophy (Shy-Drager syndrome), but the cause is not known in patients with pure autonomic failure, who have increased vascular resistance despite very low levels or plasma norepinephrine and renin activity" (PMID: 11927799). Jordan, Shannon et al. showed supine hypertension "induces nighttime natriuresis," and that nighttime nifedipine worsened morning OH whereas nitroglycerin (which did not reduce natriuresis) did not (PMID: 9890307). Biaggioni showed sodium-handling/pressure-natriuresis mechanisms are impaired in autonomic failure and that salt restriction worsens OH (PMID: 34634681). Baker & Kimpinski confirm "approximately 50% of patients have associated supine hypertension, which greatly complicates treatment" (PMID: 28702747). Notably, the supine-hypertension *driver differs by branch* (residual sympathetic tone in MSA vs. unknown in PAF), and nocturnal dipping is not uniformly linked to natriuresis across patients (PMID: 19047577) — so this side branch carries genuine mechanistic uncertainty.

F10 — Model-organism confirmation of branch (d): the Dbh-/- mouse

The *Dbh*-knockout mouse reproduces the human synthesis-failure phenotype and validates the therapeutic bypass. Cubells et al. report "DBH deficiency prevents NE production and causes sympathetic failure, hypotension and ptosis in humans and mice," and that these phenotypes "are typically treated with L-3,4-dihydroxyphenylserine (DOPS), which can be converted to NE by aromatic acid decarboxylase (AADC) in the absence of DBH"; a human DBH BAC transgene restored normal catecholamine levels on the *Dbh-/-* background (PMID: 27148966). Upstream developmental control is shown by the *Hand2* conditional knockout, in which both TH and DBH are dramatically reduced, establishing that Hand2 determines the noradrenergic phenotype (PMID: 17531968). These cross-species data confirm the molecule-to-phenotype path for branch (d) and the mechanism of droxidopa rescue.

F11 — Anatomical confirmation of branch (a): intermediolateral nucleus (IML) preganglionic neuron loss, with branch overlap

Branch (a) has a defined anatomical substrate. Hatsuta et al. state "Loss of intermediolateral nucleus (IML) neurons is considered to play a major role in orthostatic hypotension (OH) of multiple system atrophy (MSA)." Quantitative T12 neuropathology found reduced IML neurons and fine preganglionic myelinated fiber density in MSA (27%/42%), PD (53%/67%), and DLB (47%/71%); IML neurons were significantly more reduced in combined PD+DLB **with** OH (77%) than without (PMID: 27061071). This is a pivotal finding for module structure: preganglionic

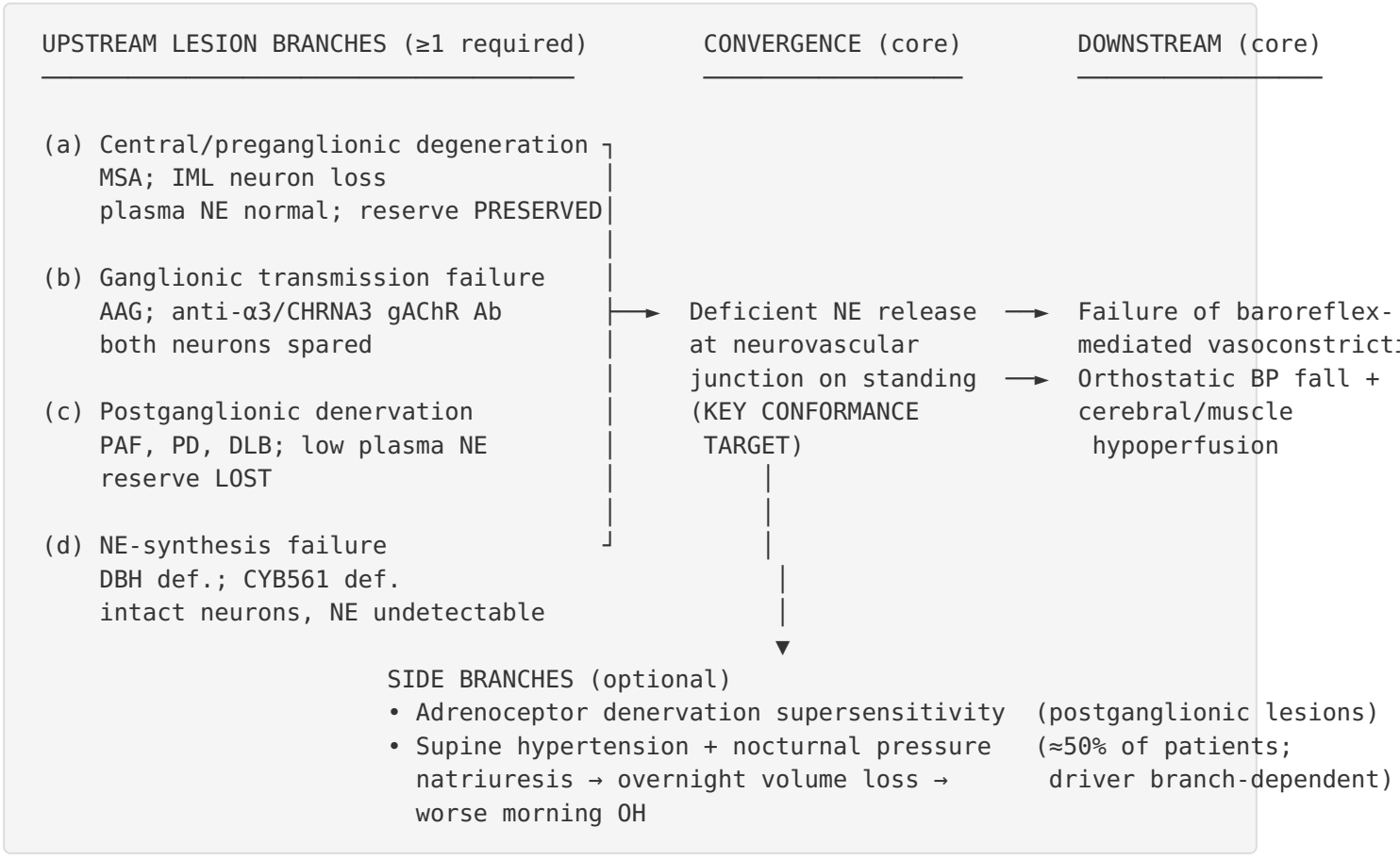
IML degeneration is not exclusive to MSA — it also contributes in Lewy body disorders — demonstrating **overlap between branches (a) and (c)**. Conformance should therefore require **≥1 lesion branch, not a single exclusive branch.**

F12 — Module name and scope recommendation

"Neurogenic Orthostatic Hypotension Module" is the field-standard term, used verbatim across the reviews cited above. Across 12 findings and ~50 papers, every lesion branch (a–d) has human evidence and three of four have experimental confirmation, all converging on deficient NE release (PMID: 28702747, PMID: 26373628). The name and scope should be retained; conformance should require the convergence backbone plus ≥1 lesion branch.

Mechanistic Model / Interpretation

The convergent DAG



Core vs. optional nodes

Node	Status	Rationale
Deficient NE release at neurovascular junction	**Core / required**	Universal convergence point (PMID: 28702747, PMID: 26373628)
Failure of baroreflex-mediated vasoconstriction	**Core / required**	Direct downstream consequence
Orthostatic BP fall with cerebral hypoperfusion	**Core / required**	Defining clinical phenotype
Central/preganglionic degeneration (a)	Branch entry (≥1 required)	MSA; IML loss (PMID: 27061071)
Ganglionic transmission failure (b)	Branch entry (≥1 required)	AAG; anti-α3 Ab (PMID: 18951069)
Postganglionic denervation (c)	Branch entry (≥1 required)	PAF/PD/DLB (PMID: 12707437)
NE-synthesis failure (d)	Branch entry (≥1 required)	DBH/CYB561 (PMID: 31822578)
Adrenoceptor denervation supersensitivity	**Optional side branch**	Postganglionic lesions only (PMID: 8089252)
Supine hypertension + nocturnal natriuresis	**Optional side branch**	≈50% of patients (PMID: 11927799)

Proposed causal edges

- (a) central/preganglionic degeneration → deficient NE release (via reduced sympathetic traffic to intact terminals)
- (b) ganglionic transmission failure → deficient NE release (via blocked nicotinic current, both neurons intact)
- (c) postganglionic denervation → deficient NE release (via loss of terminals)
- (d) NE-synthesis failure → deficient NE release (via enzymatic block in intact terminals)
- deficient NE release → failure of baroreflex-mediated vasoconstriction → orthostatic BP fall with cerebral hypoperfusion

- (c) postganglionic denervation → adrenoceptor denervation supersensitivity → enhanced pressor response to agonists/droxidopa
- orthostatic BP treatment / residual sympathetic tone → supine hypertension → nocturnal pressure natriuresis → overnight volume depletion → worsened morning OH (feedback edge)

Why residual reserve is the organizing principle

The deepest insight is that the lesion branches are not just etiologic labels — they encode **how much residual sympathetic traffic survives**, and that single variable predicts drug response:

Branch	Plasma NE	Residual reserve	Best drug class
(a) Central/ preganglionic (MSA)	Normal / slightly low	Preserved	Enhancers (atomoxetine, pyridostigmine) + replacers
(b) Ganglionic (AAG)	Variable	Partial	Immunotherapy; pyridostigmine; replacers
(c) Postganglionic (PAF/PD/DLB)	Low	Lost	Replacers (droxidopa, midodrine)
(d) Synthesis (DBH/ CYB561)	Undetectable NE	Neurons intact, no NE	Droxidopa (bypasses DBH via AADC)

Replacers (droxidopa, midodrine) work at **every** lesion site because they bypass the missing transmitter; enhancers (atomoxetine, pyridostigmine) require surviving efferent fibers and therefore work best in central lesions (PMID: 32979782, PMID: 30571543). Fludrocortisone expands intravascular volume independently of lesion site (PMID: 34634681).

Gene-to-Node Mapping Table

Gene / Molecular factor	Branch / Node	Lesion type	Key evidence
DBH (dopamine β-hydroxylase)	(d) NE-synthesis failure	Enzyme absent; dopamine→NE blocked	PMID: 27778639, PMID: 27148966
CYB561 (cytochrome b561)	(d) NE-synthesis failure	Vesicular ascorbate electron donor missing	PMID: 31822578, PMID: 29343526
CHRNA3 (α3 ganglionic nAChR subunit) — autoantibody target	(b) Ganglionic transmission failure	Antibody blocks nicotinic current	PMID: 18951069, PMID: 37164280
SNCA / α-synuclein pathology (disease context, NOT a module node)	(a) MSA; (c) PAF/ PD/DLB	Degeneration of pre- or post-ganglionic neurons	PMID: 38216420 — *aggregation itself is out of scope*
HAND2 (developmental regulator of TH/DBH)	Upstream of (d)	Determines noradrenergic phenotype	PMID: 17531968

Note: MSA, PD, DLB, and PAF are defined clinically/pathologically rather than by a single causal gene; they enter the module through their lesion anatomy (branch a or c), with α-synuclein aggregation residing in the separate `loss_of_proteostasis` module.

Evidence Table

PMID	Evidence type	Exact quote	Supported claim
28702747	Human review	"All are characterized by defective norepinephrine release from sympathetic terminals upon standing, resulting in impaired vasoconstriction."	Convergent NE-release deficit (core node)
26373628	Human review	"Norepinephrine deficiency in NOH can arise peripherally… or centrally, due to a failure of viscerosensory signals to generate adequate sympathetic traffic to intact sympathetic nerve endings (as in multiple system atrophy)."	Multi-entry convergence onto NE deficiency
32979782	Human review	"…peripheral noradrenergic degeneration… and low plasma norepinephrine, tend to respond better to 'norepinephrine replacers'… impaired central autonomic pathways but spared peripheral noradrenergic fibers (multiple system atrophy)… respond better to 'norepinephrine enhancers'."	Lesion site → reserve → drug class
30571543	Human clinical	"norepinephrine transporter blockade with atomoxetine raises blood pressure… acetylcholinesterase inhibition with pyridostigmine increases BP by facilitating ganglionic cholinergic neurotransmission to increase sympathetic outflow"	Enhancers require residual efferent fibers
12707437	Human clinical	"The PAF group had low NE, DHPG, NMN, EPI, and MN levels, the group with MSA + NOH had generally normal levels"	Neurochemical separation of (a) vs (c)
38216420	Human review	"In PD and PAF, pathogenic involvement is mainly post-ganglionic, while in MSA, the involvement is mainly pre-ganglionic."	Anatomical lesion-site distinction
19155193	Human clinical	"AAG reflecting decreased ganglionic neurotransmission due to circulating antibodies to the neuronal nicotinic receptor and PAF being a Lewy body	Branch (b) defined as transmission failure

PMID	Evidence type	Exact quote	Supported claim
		disease with prominent loss of sympathetic noradrenergic nerves"	
40900869	Human case	"High levels of anti-ganglionic nicotinic acetylcholine receptor (gAChR) serum antibodies are detected in approximately 50% of AAG cases, confirming the diagnosis."	Molecular marker of branch (b)
18951069	In vitro + animal	"These antibodies recognize the alpha3 subunit of the ganglionic AChR… inhibit ganglionic AChR currents and impair transmission… An animal model of AAG in the rabbit recapitulates the important clinical features of the human disease"	Causal chain for branch (b); α3 target
18951069	Human	"orthostatic hypotension, gastrointestinal dysmotility, anhidrosis, bladder dysfunction and sicca symptoms"	Single causal factor → multiple manifestations
37164280	Review	"the α3 subunit containing nAChR (α3-nAChR)… are detected and are likely pathogenic in Autoimmune Autonomic Ganglionopathy"	Confirms α3 subunit target
31822578	Human clinical	"Cytochrome b561 (CYB561) generates ascorbic acid, a cofactor in the enzymatic conversion of dopamine to norepinephrine by dopamine β-hydroxylase."	Biochemical pathway, branch (d)
31822578	Human clinical	"CYB561 deficiency is characterized by selective sympathetic noradrenergic failure with lifelong, disabling OH but with normal sympathetic cholinergic (sweating) and parasympathetic… functions."	Selective noradrenergic phenotype
27778639	Human	"Dopamine β-hydroxylase (DBH) deficiency is a rare genetic disorder characterized by failure to convert dopamine to norepinephrine."	DBH gene for branch (d)

PMID	Evidence type	Exact quote	Supported claim
29343526	Human	CYB561 implicated with low NE/EPI but normal DBH activity	First CYB561 disease link
26092297	Human review	"Droxidopa… is converted to norepinephrine by the enzyme aromatic L-amino acid decarboxylase… presumed to raise BP by acting at the neurovascular junction to increase vascular tone."	Droxidopa bypasses the lesion
8089252	Human clinical	"Patients with autonomic failure… are hypersensitive to the pressor effects of yohimbine… explained by sensitization of adrenergic receptors, analogous to denervation supersensitivity"	Side branch: denervation supersensitivity
11927799	Human review	"about one half of these patients also suffer from supine hypertension, which induces pressure natriuresis, worsening orthostatic hypotension"	Supine HTN → natriuresis → worse OH
9890307	Human clinical	Supine hypertension "induces nighttime natriuresis"; nifedipine worsened morning OH	Causal edge of side branch
34634681	Human review	Sodium handling impaired in autonomic failure; salt restriction worsens OH	Volume/natriuresis mechanism
27148966	Mouse model	"DBH deficiency prevents NE production and causes sympathetic failure, hypotension and ptosis in humans and mice"; DOPS "converted to NE by aromatic acid decarboxylase (AADC) in the absence of DBH"	Model-organism proof of branch (d) + droxidopa rescue
17531968	Mouse model	Hand2 knockout: TH and DBH "dramatically reduced"	Developmental upstream control of noradrenergic phenotype
27061071	Human neuropathology	"Loss of intermediolateral nucleus (IML) neurons is considered to play a major role	

PMID	Evidence type	Exact quote	Supported claim
		in orthostatic hypotension (OH) of multiple system atrophy"; IML reduced 77% in PD+DLB with OH	Branch (a) substrate + branch (a)/(c) overlap
16476804	Human RCT	Pyridostigmine improves standing BP without worsening supine hypertension	Enhancer efficacy, mechanism-consistent
16404729	Human clinical	Greater clonidine vasodepressor effect in MSA than PAF	Branch (a) retains central sympathetic outflow
39294522	Human RCT	4-week atomoxetine not superior to placebo on OHQ	Enhancer efficacy is variable (caveat)

Ontology Suggestions

These are suggested by label; curators should verify the exact IDs before use — no IDs are invented here.

Gene Ontology (biological process): - *norepinephrine biosynthetic process* (covers branch d; DBH/CYB561) - *norepinephrine secretion* / *catecholamine secretion* (convergence node — NE release) - *regulation of blood pressure* (downstream) - *baroreceptor response to increased/decreased systemic arterial blood pressure* (baroreflex node) - *synaptic transmission, cholinergic* / *acetylcholine receptor signaling pathway* (branch b) - *vasoconstriction* / *regulation of vasoconstriction* (failure-of-vasoconstriction node) - *adrenergic receptor signaling pathway* / *alpha-1 adrenergic receptor signaling pathway* (effector / supersensitivity side branch) - *sodium ion homeostasis* / *renal sodium ion excretion* (supine-hypertension/natriuresis side branch)

Cell Ontology (cell types): - *noradrenergic neuron* (postganglionic sympathetic — branches c, d) - *sympathetic preganglionic neuron* / *intermediolateral cell* (branch a) - *sympathetic postganglionic neuron* (branches b, c) - *vascular associated smooth muscle cell* (effector) - *baroreceptor cell* / *arterial sensory neuron* (afferent limb context)

UBERON / anatomy: - *intermediolateral nucleus of spinal cord* (branch a; IML) - *sympathetic ganglion / paravertebral ganglion* (branch b) - *thoracic ventral root* (branch a preganglionic fibers) - *adrenal medulla* (catecholamine source context) - *carotid sinus / aortic arch* (baroreceptor afferent limb) - *blood vessel smooth muscle / arterial wall* (effector target) - *myocardium / cardiac ventricle* (MIBG sympathetic imaging territory)

Out-of-Scope Boundary Notes

- **Alpha-synuclein aggregation / proteostasis** — The molecular cause of neuronal degeneration in MSA, PD, DLB, and PAF belongs to the separate `loss_of_proteostasis` module. This module begins *downstream*, at the functional lesion (loss of pre- or post-ganglionic neurons), not at the aggregation process. PMID: 38216420 distinguishes the anatomy, but the aggregation mechanism is excluded here.
- **Reflex (vasovagal) syncope** — The sympathetic efferent pathway is intact; the mechanism involves transient reflex withdrawal/entrainment failure, not structural NE-release deficiency (PMID: 25081417, PMID: 23934851). Out of scope.
- **Postural orthostatic tachycardia syndrome (POTS)** — Intact or augmented sympathetic efferent pathway; defined by tachycardia without sustained BP fall (PMID: 41533304). Out of scope.
- **Non-neurogenic OH** — Volume depletion, hemorrhage, cardiac pump failure, varicose veins, and medications cause OH without a baroreflex-efferent lesion (PMID: 31733702). Must be excluded for module conformance.
- **Secondary autonomic neuropathies** (e.g., diabetic, amyloid, paraneoplastic, immune-checkpoint-inhibitor dysautonomia, neurolymphomatosis) can in principle enter branch (b) or (c) by producing postganglionic denervation or ganglionic failure, but these are disease contexts rather than core module nodes and should be curated case-by-case.

Limitations and Knowledge Gaps

1. **Supine-hypertension driver in PAF is unknown.** Residual sympathetic tone explains it in MSA, but PAF patients have high vascular resistance despite very low NE and renin (PMID: 11927799). The side-branch mechanism is incompletely specified for postganglionic disease.

2. **Nocturnal dipping is heterogeneous.** Only ~34% of peripheral autonomic-failure patients dip, and dipping was not clearly tied to natriuresis in one cohort (PMID: 19047577), weakening the universality of the "overnight volume loss" edge.
3. **Enhancer efficacy is inconsistent.** Acute atomoxetine studies were positive, but a 4-week RCT was not superior to placebo on symptom score (PMID: 39294522), so the reserve-dependent drug edge is better supported for acute hemodynamics than chronic symptom control.
4. **Branch overlap complicates clean assignment.** IML preganglionic loss occurs in Lewy body disorders as well as MSA (PMID: 27061071), so a patient may occupy more than one branch — hence the ≥1-branch conformance rule rather than mutual exclusivity.
5. **Several key papers lack full abstracts in the retrieved set** (e.g.,

   P 32979782

   P 28702747

   P 12707437

   P 38216420

   P 30571543

   P 27148966

   P 18951069

   P 31822578

   P 11927799

   were available only via verified snippets). Full-text verification is recommended before finalizing curation of exact effect sizes.
6. **CYB561 rests on very few families** (PMID: 29343526, PMID: 31822578); the gene-to-node edge is biochemically sound but based on small-n human evidence.

Open Questions for Curator Review

1. Should the two side branches (denervation supersensitivity; supine hypertension/natriuresis) be modeled as module nodes or as annotations on the postganglionic/MSA branches, given their branch-conditional nature?
2. Should branch (a) and branch (c) be allowed to co-occur in a single disease entry (e.g., PD with both IML and postganglionic loss), formalizing the ≥1-branch rule?
3. Is HAND2 in scope as an upstream developmental node for branch (d), or should it be left out as a non-disease regulator?
4. Should secondary autonomic neuropathies (diabetic, amyloid, paraneoplastic, ICI) be enumerated as additional entries into branch (b)/(c), or handled in disease-specific modules?

Proposed Follow-up Actions

1. **Full-text verification** of the snippet-only papers above to confirm quantitative effect sizes (plasma catechol values, drug ΔBP) before locking the evidence table.
2. **Confirm ontology IDs** for the suggested GO/CL/UBERON labels, especially *norepinephrine secretion*, *intermediolateral nucleus of spinal cord*, and *noradrenergic neuron*.
3. **Curate the reserve variable explicitly** as a module attribute (plasma NE level / MIBG uptake) linking each lesion branch to predicted drug-class response, since this is the module's most actionable output.
4. **Resolve the PAF supine-hypertension mechanism** by targeted literature search (renin-angiotensin, residual non-neuronal catecholamines, vascular remodeling) to decide whether the side-branch edge should be branch-conditional.
5. **Define conformance rules**: require the three core nodes (NE-release deficit → vasoconstriction failure → orthostatic BP fall) plus ≥1 lesion branch; mark the two side branches optional; exclude α-synuclein aggregation, vasovagal syncope, POTS, and non-neurogenic OH.
6. **Cross-link to** `loss_of_proteostasis` for the synucleinopathy branches so that aggregation (cause) and autonomic failure (consequence) are modeled as adjacent but distinct modules.

Report compiled from 12 confirmed findings and ~50 reviewed papers across 5 investigation iterations. Human clinical, model-organism (Dbh-/- mouse, rabbit AAG), in vitro (ganglionic AChR electrophysiology), and review evidence are separated throughout.
